11-08-2026

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Experiment 4:

AIM: Write a python program to implement Simple Linear Regression for House Price Prediction

PROGRAM:

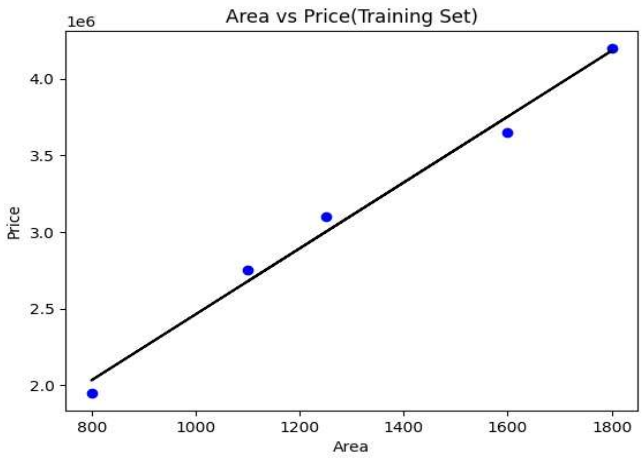
```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
from sklearn import linear_model data=pd.read_csv("C:/Users/MPIT/Downloads/house_price.csv”)
data
x=data.iloc[:, :-1].values
x
y=data.iloc[:,1].values
y
from sklearn.model_selection import train_test_split
X_train,X_test,y_train,y_test=train_test_split(x,y,test_size=0.3,random_state=0)
X_train
X_test
from sklearn.linear_model import LinearRegression
regressor=LinearRegression()
regressor.fit(X_train,y_train)
y_pred=regressor.predict(X_test)
y_pred
y_test
plt.scatter(X_train,y_train,color='blue')
plt.plot(X_train,regressor.predict(X_train),color='black')
plt.title("Area vs Price(Training Set)")
plt.xlabel("Area")
plt.ylabel("Price")
plt.show()
plt.scatter(X_test,y_test,color='blue')
plt.plot(X_train,regressor.predict(X_train),color='black')
plt.title("Area vs Price(Test Set)")
plt.xlabel("Area")
plt.ylabel("Price")
plt.show()
print("Coeffecient:",regressor.coef_)
print("Intercept:",regressor.intercept_)
p=regressor.predict(np.array([[4500]]))
print("Predicted Price=",p[0])
```

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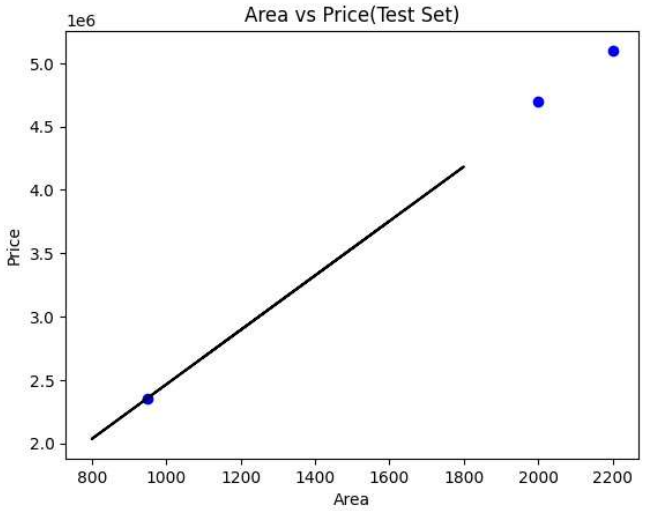
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Output:

```
   Area_sqft  Price
0    1800      4200000
1     950      2350000
2    2200      5100000
3    1250      3100000
4    1600      3650000
5    800       1950000
6    2000      4700000
7    1100      2750000
array([[1800],
       [ 950],
       [2200],
       [1250],
       [1600],
       [ 800],
       [2000],
       [1100]], dtype=int64)
array([4200000, 2350000, 5100000, 3100000, 3650000, 1950000, 4700000, 2750000], dtype=int64)
array([[1100],
       [1250],
       [1800],
       [ 800],
       [1600]], dtype=int64)
array([[2000],
       [2200],
       [ 950]], dtype=int64)
array([4613172.46835443, 5043077.53164557, 2356170.88607595])
array([4700000, 5100000, 2350000], dtype=int64)
```



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Coeffecient: [2149.52531646]
Intercept: 314121.8354430385
Predicted Price= 9986985.75949367

Mean Absolute Error for Simple Linear Regression:

PROGRAM:

```
from sklearn.metrics import mean_absolute_error,mean_squared_error
mae=mean_absolute_error(y_test,y_pred)
mae
```

Output:

49973.62869198341

Mean Squared Error for Simple Linear Regression:

PROGRAM:

```
mse=mean_squared_error(y_test,y_pred)
mse
```

Output:

3605755830.061998

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Root Mean Squared Error for Simple Linear Regression:

PROGRAM:

```
rmse=np.sqrt(mse)
rmse
```

Output:

60047.94609361754